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Cryptography & Network Security

Security -

The state of being safe or free from potential damage, loss or changes.

Threat

An entity like a person or an Object or an action that may cause harm, damage, loss or changes to a system.

Vulnerability

It is a loophole or weakness in a system which can be exploited by the attackers to enter into the system causing harm, damage or loss to the system.

Attack

An attack is an action by the malicious attackers to exploit the vulnerability & causing actual damage, harm or loss to the system.

Security Goals (Security traids)

3 security goals: (CIA)

i) Confidentiality
ii) Integrity
iii) Availability

Cryptography –

The art & science of sending data in a secret manner.

Encryption

The process of transforming data in intelligent form to an unintelligent form.

Decryption

The process of converting an unintelligent data back to intelligent form

Plain text

The original message to be transmitted is known as a plain text (P).

Cipher text

The message obtained after encryption is known as cipher text. (C).

Key (K)

A value used for encryption & decryption.

Cipher

It refers to all encryption & decryption algorithms

Encryption –

$$C = E_K(P)$$

$$E_K(P) = (P+K) \bmod 26$$

Decryption

$P = D_k(C)$

$D_k(C) = (C - k) \bmod 26$

Components of computer network

Sender - Desktop, laptop, sensor, mobile

Receiver -

Data / message

Communication media - wired media (coaxial cable, twisted wire pair, optical fibre)

wireless media (microwave, radio wave, satellite, infrared)

Protocols - sets of rules / conventions

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Cryptology -

The branch of science that deals with encryption & decryption, key management etc.

Cryptanalyst

A person who can break a ciphertext without the given key & can exploit the security system to check strength of security system.

Modular arithmetic

Arithmetic where the operation is modulus operation (to find remainder)

$r = a \bmod n$

→ It is used to map the result of any operation into a particular set.

Properties

1. $(a \pm b) \bmod n = (a \bmod n \pm b \bmod n) \bmod n$

$(37 + 25) \bmod 4$

$62 \bmod 4 = 2$ $= 37 \bmod 4 + 25 \bmod 4$

$= 1 + 1 = 2 \bmod 4$

2. $(a \times b) \bmod n = (a \bmod n \times b \bmod n) \bmod n$

$(25 \times 30) \bmod 7 = (25 \bmod 7 \times 30 \bmod 7) \bmod 7$

$= (4 \times 2) \bmod 7 = 1 \bmod 7$

$1 \bmod 7$

3. **Additive inverse**

Two elements a & b w.r.t mod n are said to be additive inverse of each other if $(a+b) \bmod n = 0 \bmod n$

4 Multiplicative inverse

Two elements a & b w.r.t mod n are said to be multiplicative inverse of each other if $(a \times b) \bmod n = 1 \bmod n$

2 & 4 are multiplicative inverse of each other w.r.t mod 7

Important sets in modular arithmetic

Z_n = set of all integers ~~when~~ as a result of mod n.
= Residual set.

Eg: $Z_5 = \{0, 1, 2, 3, 4\}$ set of all remainders / residue when any value is divided by 5.

$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Additive inverse pairs in Z_{10} are
(0,0), (1,9), (2,8), (3,7), (4,6), (5,5)

Multiplicative inverse pairs in ~~Z_{10}~~ Z_{10}
= (1,1), (3,7), (7,3), (9,9)

$Z_n = \{0, 1, 2, \ldots, n-1\}$

Z_n^* = Set of all values in Z_n having a multiplicative inverse

i.e the values in Z_n which are coprime to n.

$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$Z_{10}^* = \{1, 3, 7, 9\}$

$Z_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$

$Z_8^* = \{1, 3, 5, 7\}$

* Additive inverse of all values in Z_n exist but multiplicative inverse of all values in Z_n^* doesn't exist.

Z_p = set of all ~~residue~~ residues w.r.t mod p where p is a prime.

Eg: $Z_7 = \{0, 1, 2, 3, 4, 5, 6\}$

Z_p^* = set of all values in Z_p whose multiplicative inverse exist

$Z_7^* = \{1, 2, 3, 4, 5, 6\}$

Addition table of Z_5

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Multiplication table of $\mathbb{Z}_4$

X	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

Congruent relation in $\mathbb{Z}_n$

a & b are congruent to each other if $a \bmod n = b \bmod n$, it is denoted as $a \equiv b \bmod n$,

$9 \equiv 16 \bmod 7$

$9 \equiv 2 \bmod 7$

$n = 7$

$9 \bmod 7 = 2$

$16 \bmod 7 = 2$

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Two conditions / constraints of modular arithmetic

1. The divisor n can not be zero (0)
2. The remainder is non-negative.

$-18 \bmod 12 = -6 + 12 = 6$

$-35 \bmod 12$

$12 \mid -35 \mid -2$
-24
$+$
-11

$12 \mid -35 \mid -3$
-36
$+$
1

Euclidean Extended Algorithm to find multiplicative inverse of a value in Z_n

Euclidean Algorithm

1. $Gcd(a,0) = a$
2. $Gcd(a,b) = gcd(b, r)$ where $r = a \bmod b$

HCF

Eg: $gcd(\overset{a}{36}, \overset{b}{14})$

$= gcd(14, 8)$
$= gcd(8, 6)$
$= gcd(6, 2)$
$= gcd(2, 0) = 2$

Extended Euclidean Algo. to find a^{-1} in Z_n

$r_1 = n$; $r_2 = a$
$t_1 = 0$; $t_2 = 1$
while $(r_2 > 0)$
{
$q = r_1 / r_2$
$r = r_1 - q * r_2$

```
r1 = r2; r2 = r;
t = t1 - q x t2
t1 = t2; t2 = t;
}
if r1 == 1
return t1 as MI of a
```

Q Find MI of 7 in Z_{10}

$r_1 = 10, 7, 3, 1$ $r_2 = 7, 3, 1, 0$ $t_1 = 0, 1, -3$ $t_2 = 1, -1, 3, -10$

$q = 1, 2, 3$ $r = 3, 1, 0$ $t = -1, 3, -10$

MI of 7 = 3

Q Find MI of 13 in Z_{26}

$r_1 = 26, 13$, $r_2 = 13, 0$, $t_1 = 0, 1$, $t_2 = 1, -2$

$q = 2$ $r = 0$ $t = -2$

Q Find MI of 15 in Z_{26}

$r_1 = 26, 15, 11, 4, 3, 1$ $r_2 = 15, 11, 4, 3, 1, 0$ $t_1 = 0, 1, -1, 2, -5, 7, -26$ $t_2 = 1, -1, 2, -5, 7, -26$, $q = 1, 1, 2, 1, 3$ $r = 11, 4, 3, 1, 0$

MI of 15 = 7

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Linear Congruent Equation

$a \equiv b \bmod n$

$a \bmod n = b \bmod n$

An equation of the form

$ax \equiv b \bmod n$

is known as a linear congruent equation.

Suppose $\gcd(a,n) = d$

if $d \mid b$ (d divides b) then there are d sol^{n}s.

if $d \nmid b$ (d doesnot divide b) then there are no sol^{n}s.

Steps to solve linear congruence eq^n.

1. find $d = \gcd(a,n)$
2. divide both sides by d including mod.

$$\frac{ax}{d} \equiv \frac{b}{d} \bmod \frac{n}{d}$$

3. Multiply the RHS with multiplicative inverse of a to find the particular sol^n x_0.
4. The general sol^n is

$$x = x_0 + k\left(\frac{n}{d}\right) \quad k = 0, 1, 2 \ldots, d-1$$

Eg:- 1. Solve $4x \equiv 6 \bmod 8$

$d = \gcd(4, 8) = 4$

Since 4 doesn't divide 6, there is no sol^n.

2. Solve $\underset{ax}{2x} \equiv \underset{b}{6} \bmod \underset{n}{8}$

$d = \gcd(2, 8) = 2$

$\frac{2x}{2} \equiv \frac{6}{2} \bmod \frac{8}{2}$

$\Rightarrow x \equiv 3 \bmod 4$

$x_0 = 3$

$x_1 = x_0 + 1 \times \frac{8}{2} = 3 + 4 = 7$

Sol^n = (3, 7)

3. Solve $14x \equiv 12 \bmod 18$

$d = \gcd(14, 18) = 2$

$\frac{14}{2} x \equiv \frac{12}{2} \bmod \frac{18}{2}$

$7x \equiv 6 \bmod 9$

$x \equiv \frac{6}{7} \bmod 9$

$x \equiv 6 \times 7^{-1} \bmod 9$

$x = 6 \times 4 \bmod 9$

$x = 24 \bmod 9$

7^{-1} in mod 9 means 7 ko kis no. se multiply krenge jo jb 9 se divide hoga tb rem. 1 aayega

$x_1 = 6 + 1 \times \frac{18}{2} \mod 18$

~~$= 6 \mod 18 = 6 \mod 18$~~

$= 15 \mod 18$

{6, 15}

Traditional symmetric key cipher

Two types of encryption techniques

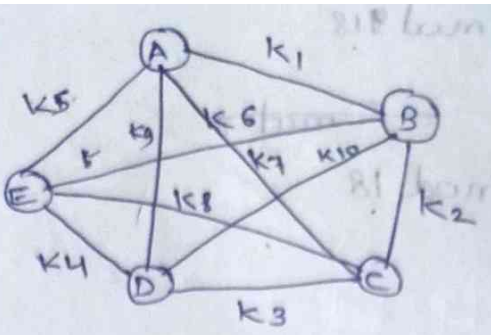
1) Symmetric key / private key / shared key encryption / secret key encryption

2) Asymmetric key / public key encryption

Symmetric key / shared key	Asymmetric key / public key encryption
1. A private / secret key is shared b/w sender & receiver.	1. A pair of key is used for encryption & decryption.
2. The same key is used for encryption as well as decryption.	2. Public key is used for encryption & private key is used for decryption.
3. The no. of keys is increased as no. of entities is also increased.	3. It uses less no. of keys as compared to symmetric key encryption.

n = no. of entities

Total no. of secret key $= \frac{n(n-1)}{2}$



$\frac{n(n-1)}{2}$

$\frac{5 \times 4}{2} = 10$

$n = 100$

$\frac{n(n-1)}{2} = \frac{100 \times 99}{2} = 4950$

Traditional symmetric key Cryptography

Symmetric key ciphers

- Stream cipher
  - MonoAlphabetic
  - PolyAlphabetic
- Block cipher

Symmetric ciphers

- Substitution cipher
- Transposition Cipher

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In shared key / symmetric key cryptography, encryption & decryption are reverse of each other.

cryptography
- shared key / symmetric key / private key encryption
- Assymetric key / public key cryptography

Symmetric key cipher
- Stream cipher (only one character is encrypted at a time)
- Block cipher (A group of characters are encrypted at a time).

Traditional symmetric key cipher

i) Substitution cipher
ii) Transposition cipher

Monoalphabetic substitution cipher

→ Every alphabet in plain text has exact one substitution alphabet i.e ciphertext.
→ The ~~translation~~ relation between plaintext & ciphertext character is one to one.
→ For all ciphers, the residue set is Z_{26}.
→ The plain text alphabets are represented in lowercase
→ Ciphertext alphabets are represented in upper case.

Eg:- Additive cipher / shift cipher

Encryption

$C = E_k(P)$

$E_k(P) = (P+k) \bmod 26$

Eg:- P = "manager"

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 ... 25
a b c d e f g h i j k l m n ... z

k = 4

$C_1 = (12+4) \bmod 26 = 16$

$= Q$

$C_2 = (0+4) \bmod 26 = 4$

$= E$

$C_3 = (13+4) \bmod 26 = 17$

$= R$

$C_4 = E$

$C_5 = (6+4) \bmod 26 = 10$

$= K$

$C_6 = (4+4) \bmod 26 = 8$

$= I$

$C_7 = (17+4) \bmod 26 = 21$

$= V$

The cipher text C = "QERE K I V"

Decryption

$$P = D_k(C)$$

$$D_k(C) = (C - K) \bmod 26$$

Eg of additive cipher is Caesar cipher.

Multiplicative cipher

Encryption

$$E_k(P) = (P \times K) \bmod 26$$

$$D_k(C) = (C \div K) \bmod 26$$

$$= (C \times K^{-1}) \bmod 26$$

Constraint

→ The values in Z_{26} can be considered to be a key in multiplicative cipher, provided their multiplicative inverse in Z_{26} exists.

Eg: P = "Captain"

K = 15

$C_1 = (2 \times 15) \bmod 26$
$= 4 = E$

$C_2 = (0 \times 15) \bmod 26$
$= 0 = A$

$C_3 = (15 \times 15) \bmod 26$
$= 17 = R$

$C_4 = (19 \times 15) \bmod 26$
$= 25 = Z$

$C_5 = 0 = A$

$C_6 = (8 \times 15) \bmod 26$
$= 16 = Q$

$C_7 = (13 \times 15) \bmod 26$

$= 195 \bmod 26$

$= 13 = N$

$C = EARZAQN$

$P_1 = (4 \times 7) \bmod 26 = 2 = c$

$P_2 = (0 \times 7) \bmod 26 = 0 = a$

$P_3 = (17 \times 7) \bmod 26 = 15 = p$

$P_4 = (25 \times 7) \bmod 26 = 19 = t$

$P_5 = (0 \times 7) \bmod 26 = 0 = a$

$P_6 = (16 \times 7) \bmod 26 = 8 = i$

$P_7 = (13 \times 7) \bmod 26 = 13 = n$

P = captain

$Z_{26}^* = \{1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25\}$

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Affine Cipher

→ It combines both multiplicative & additive cipher.

→ More robust than additive & multiplicative cipher individually.

Encryption

$T = (P \times K) \bmod 26$

$C = (T + K) \bmod 26$

Decryption

→ $T = (C - K) \bmod 26$

$P = (T + K) \bmod 26$

Hlw

Q) Encrypt the plain text "attack" by using affine cipher with K = 17.

Polyalphabetic substitution cipher

→ An alphabet in plain text can be substituted by multiple alphabets in cipher text.

→ The relation b/w plain text & cipher text alphabets is one-to-many.

Eg:- Auto Key Cipher

→ The first key in the cipher is a secret key

→ The 2nd key is the 1st alphabet of plain text.

→ The 3rd key is the 2nd alphabet of plain text.

$P = P_1\ P_2\ P_3\ P_4\ P_5\ P_6$

$K = K_1\ K_2\ K_3\ K_4\ K_5\ K_6\ K_7$

↙ Secret key

Encryption

$$C_i = (P_i + K_i) \bmod 26$$

Q Encrypt the message "enemy attacks tonight" by using auto key cipher with initial key (K) as 4.

P = e n e m y a t t a c k s t o n i g h t
04 13 04 12 24 00 19 19 00 02 10 18 19 14 13 08 07 19
08 02
06

K = 04 04 13 04 12 24 00 19 19 00 02 10 18 19 14 13 08

C = 08 17 17 16 10 24 19 12 19 02 12 02 11 07 01 21 14 13

I = R R Q K Y T M T C M C L H B V O N A

Playfair cipher

→ It is a polyalphabetic substitution cipher used by British Army during World War-I

→ The key used in playfair cipher is a 5X5 matrix of random alphabets where i & j belong to same cell in the matrix.

K =

B	L	G	Z	O
H	P	T	R	U
N	A	I/J	C	X
Q	S	E	Y	V
D	K	W	M	F

go
TW

Logo

to go
GB ZB

25!

H/W

Ans: P = a t t a c k

0 19 19 0 2 10

C = 10 | 15 | 15 | 10 | 18 | 24

k | P | P | k | S | Y

$(17 \times 0 + 10) \bmod 26 = 10$

$(19 \times 19 + 10) \bmod 26 = 15$

$(17 \times 2 + 10) \bmod 26 = 10$

$(17 \times 10 + 10) \bmod 26 = 24$

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$C = (P + 17) \bmod 26$

$C = 0) 17 \bmod 26$

$T_1 = (0 \times 17) \bmod 26 = 0$

$C_1 = (0 + 17) \bmod 26 = 17 = R$

$T_2 = (19 \times 17) \bmod 26 = 14$

$C_2 = (14 + 17) \bmod 26 = F$

$T_3 = (19 \times 17) \bmod 26 = 14$

$C_3 = (14 + 17) \bmod 26 = 5 = F$

$T_4 = (0 \times 17) \bmod 26 = 0$

$C_4 = (0 + 17) \bmod 26 = 17 = R$

$T_5 = (2 \times 17) \bmod 26 = 8$

$= (8 + 17) \bmod 26 = 25 = Z$

$T_6 = (10 \times 17) \bmod 26 = 14$

$C_6 = (14 + 17) \bmod 26 = 5 = F$

C = RFFRZF

→ Encryption

→ All plain text characters are written as pairs after remaining space.

→ An extra alphabet is appended at the end to make last pair.

→ If the pair of alphabets appear in the same row in key matrix, they are encrypted as next alphabets in the same row of the key matrix.

→ If the pair of alphabets appear in the same column, they are encrypted by the alphabet below them in the same column.

→ If a pair alphabets doesn't appear in the same row or same column, the ~~pair is encrypted~~ by first alphabet is encrypted by the alphabet that belongs to same row & the column of 2nd alphabet.

Eg: high
TN BT

Q Encrypt the message "attack tonight"

at ta ck
IP PI AM

to ni gh t
UG AC BT UI

Vigenere Cipher

P = P1 P2 P3 ...

C = C1 C2 C3 ...

K = k1 k2 k3 ... km

→ A fixed length key (m) is repeated after every m alphabets of plain text.

→ Encryption $C_i = (P_i + K_i) \bmod 26$

Decryption $P_i = (C_i - K_i) \bmod 26$

P = P1, P2, P3, P4, P5, P6, P7, P8, P9, P10, P11, P12, P13

K = K1, K2, K3, K4, K5 k1 k2 k3 k4 k5 k1 k2 k3

Q Encrypt the message "meet at river site" by using Vegenere cipher with key "PASCAL".

P: m e e t a t r i v e r s i t e

P value: 12 04 04 19 00 19 17 8 21 4 17 18 8 19 4

K value: 15 00 18 02 11 15 00 18 02 11 15 00 18 02 11

K: P A S C A L P A S C A L P A S C A

Add K & P to get C

P: m e e t a t r i v e r s i t e

P value: 12 04 04 19 00 19 17 8 21 4 17 18 8 19 4

K value: 15 00 18 02 00 11 15 00 18 02 00 11 15 00 18

C value: 01 04 22 21 00 04 06 08 13 06 17 03 23 19 22

C: B E W V A E G I N G R D X T W

Hill cipher

→ It uses matrix operation like multiplication & inverse.

→ It is a block cipher where the plain text alphabets are arranged in block of fixed size (say m).

→ The key is a m×m square matrix

$$P = \begin{bmatrix} P_{11} & P_{12} & P_{13} & \cdots & P_{1m} \\ P_{21} & P_{22} & P_{23} & \cdots & P_{2m} \\ \vdots & & & & \\ P_{n1} & P_{n2} & P_{n3} & \cdots & P_{nm} \end{bmatrix} \quad K = \begin{bmatrix} K_{11} & K_{12} & K_{13} & \cdots & K_{1m} \\ K_{21} & K_{22} & K_{33} & \cdots & K_{2m} \\ \vdots & & & & \\ K_{m1} & K_{m2} & K_{m3} & \cdots & K_{mm} \end{bmatrix}$$

Encryption $C = P^{n \times m} \times K^{m \times m}$

Decryption $P = C \times K^{-1}$

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Eg:- Encrypt the message "code is ready" by using hill cipher with key = $\begin{bmatrix} 09 & 07 & 11 & 13 \\ 04 & 07 & 05 & 06 \\ 02 & 21 & 14 & 09 \\ 03 & 23 & 21 & 08 \end{bmatrix}$

$$P = \begin{bmatrix} c & o & d & e \\ i & s & r & e \\ a & d & y & z \end{bmatrix}$$

$$= \begin{bmatrix} 02 & 14 & 03 & 04 \\ 08 & 18 & 17 & 04 \\ 00 & 03 & 24 & 25 \end{bmatrix}$$

$C = P \times K$

$$= \begin{bmatrix} 02\times09 + 14\times04 + 03\times02 + 04\times03 & - & - & - \\ 08\times09 + 18\times04 + 17\times02 + 04\times03 & - & - & - \\ 00\times09 + 03\times04 + 24\times02 + 25\times03 & - & - & - \end{bmatrix}$$

$$= \begin{bmatrix} 14 & - & - & - \\ 08 & - & - & - \\ 05 & - & - & - \end{bmatrix}$$

$P = C \times K^{-1}$

K = "JFHLNEHFGCVOJDXVI"

Transposition Cipher

→ The position of plain text alphabets are permuted to get cipher text alphabets.

→ Two types of transposition cipher:-

i) Key less transposition

ii) Keyed transposition

Keyless transposition cipher

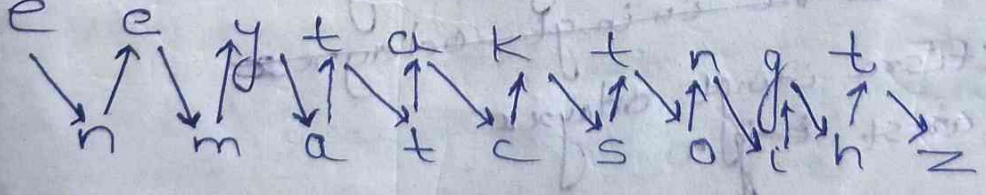
i) Arrange the plain text in columns & encrypt it row by row.

ii) Arrange the plain text in row & encrypt it column by column.

Railfence cipher

→ The plain text alphabets are written in zigzag manner (i.e two alphabets per column) & read it row by row (two rows).

Eg:- P = "enemy attacks tonight"



C = "EEYTAKTNGTNGTNMATCSOIHZ"

ii)

e n e m
g a t t
a c k s
t o n i
g h t z

C = EYATGNACOHETKNTMISIZ"

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
↓															
1	6	11	16	02	03	04	12	07	08	13	18	15	09	14	

Keyed transposition.

→ A permutation key is used to encrypt every alphabets in a block.
→ The plain text is divided into blocks of size m.

Key: E

P	1	2	3	4	5	6
C	4	6	1	5	3	2

↑ D

P = "it is raining today"

~~it is ra in ingt odayxx~~

itisra ingt odayxx

iarist itginn axxoyd

C = "IARISTITGINNAXXOYD"

~~C = "iarist itginn~~

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Algebraic Structure

Group :-

A structure $G = \langle S, \cdot \rangle$ ('Binary operation') is said to be a group if it satisfies following properties.

S = set of elements / integers

$\cdot$ = binary operator

i) Closure property :-

if $a, b \in S$, $a \cdot b \in S$

ii) Associative property :-

if $a, b, c \in S$

$a \cdot (b \cdot c) = (a \cdot b) \cdot c$

iii) Identity property :-

For every element a in set S, there exist an element e such that $a \cdot e = a$, then e is called as identity element.

iv) Inverse property :-

For every element a in set S, there exist an element b such that $a \cdot b = e$

b is inverse of a w.r.t the operator '$\cdot$'

A group is said to be commutative group or abelian group if it satisfies an extra property i.e commutative property.

$$a \cdot b = b \cdot a$$

Eg 1:- The set of integers Z is an abelian group w.r.t addition '+'.

$G = \langle Z, + \rangle$ is a group

2. $G = \langle Z, \times \rangle$ is an abelian group

False.

3. $G = \langle Z_n, + \rangle$ is an abelian group

True

4. $G = \langle Z_n, * \rangle$ is not a group as multiplicative inverse of every element in Z_n doesnot exist.

5. $G = \langle Z_n^*, * \rangle$ is a group?

$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$Z_{10}^* = \{1, 3, 7, 9\}$

Yes it is a group.

$G_1 = \langle Z_n^*, + \rangle$ is not a group.

6. Consider the set $S = \{a, b, c, d\}$ alongwith the operator '$\cdot$'. The result of operation is as follows:

$\cdot$	a	b	c	d
a	a	b	c	d
b	b	c	d	a
c	c	d	a	b
d	d	a	b	c

Verify whether $Gr = \langle S, \cdot \rangle$ is a group or not.

Yes it is a group & abelian group.

Finite group -

A group Gr is said to be a finite group if the set of elements is a finite set.

*Order of a group Gr denoted as $|Gr|$ is the no. of elements in the set S.

Subgroup

A group $H = \langle T, \cdot \rangle$ is said to be a subgroup of a group $Gr = \langle S, \cdot \rangle$ if T is a finite subset of S.

& H is a group itself.

Ex: $Z_7 = \{0, 1, 2, 3, 4, 5, 6\}$

$T = \{0, 2, 4, 5\}$

$G = \langle Z_7, + \rangle$ It is a subgroup of G.

$H = \langle T, + \rangle$

→ A group is a subgroup of itself

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Ring -

A ring denoted as $R = \langle \{\ldots\}, \cdot, \square \rangle$

Set of elements & two binary operators provided the following properties are satisfied.

→ The 1st operator satisfies all 5 properties of a group.

→ The 2nd operator must satisfy 1st two properties (closure & associative) & commutative property for a commutative ring

•

i) closure
ii) Associative
iii) Commutative
iv) Inverse
v) Identity

□

i) closure
ii) Associative
iii) commutative
(commutative ring)

→ The 2nd operator must be distribute over the 1st one.

Eg:- i) $R = \langle Z, +, \times \rangle$ is a commutative ring.

Distributive property

$a \times (b+c) = a \times b + a \times c$

ii) $R = \langle Z, \times, + \rangle$ is not a ~~commutative~~ ring.

iii) $R = \langle Z_n, +, \times \rangle$ is a ring?

It is a ring.

Field

An algebraic structure $F = \langle \{ \ldots \}, *, \square \rangle$ with a set of elements & two binary operators is said to be a field if both the operators satisfies all five properties of a group except the 2nd operator doesn't have inverse of identity element of 1st operator.

Eg:- i) $F = \langle Z, +, \times \rangle$ is a field

→ No, ~~Every element~~ ~~X doesn't~~

→ No, multiplicative inverse of every element doesn't exist in Z.

ii) $F = \langle Z_n, +, \times \rangle$ is a field?

→ No.

iii) $F = \langle Z_n^*, +, \times \rangle$ is a field?

→ ~~Yes~~ No

$Z_{10}^* = \{1, 3, 7, 9\}$

iv) $F = \langle Z_p, +, \times \rangle$ is a field

$Z_7 = \{0, 1, 2, 3, 4, 5, 6\}$

+ pass all properties

× pass all properties

Yes.

~~F = Z~~

Finite field

A field is said to be a finite field if the set of elements is finite

→ Finite fields are also known as Galloi's field (GF).

GF(P) is a finite set

where P is a prime no.

Application of fields in cryptography

GF(2^n) is field used in computer/cryptography

GF(2) $F = \langle \{0,1\}, +, \times \rangle$ $n = 1$

XOR

+	0	1
0	0	1
1	1	0

XOR operation

X	0	1
0	0	0
1	0	1

AND ~~XOR~~ operations



→ In computer words are represented in terms of n bits ($n = 16, 32, 64 \ldots$)

Eg:- GF(2^3) = words are represented in terms of 3 bits.

{000, 001, 010, 011, 100, 101, 110, 111}

GF(2^2)

$\langle \{00, 01, 10, 11\}, + \times \rangle$

⊕	00	01	10	11
00	00	01	10	11
01	01	00	11	10
10	10	11	00	01
11	11	10	01	00

XOR

X	00	01	10	11
00	00	00	00	00
01	00	01	00	01
10	00	00	10	10
11	00	01	10	11

AND

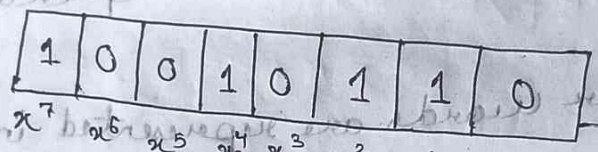
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→ A polynomial of degree n-1 is used to represent an n-bit word.

→ The coefficients are either 0 or 1.

Eg:- 8 bit word.

10010110



$a_0x^0 + a_1x^1 + a_2x^2 + \dots a_nx^n$

$= 1x^7 + 0x^6 + 0x^5 + 1x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0$

$= x^7 + x^4 + x^2 + x$

$x^6 + x^5 + x^3 + x + 1$

01101011

Irreducible Polynomial

Multiplication of two polynomial of degree n-1 may created a polynomial with degree more than n-1.

In this case we need to divide the result by a modulus polynomial. The modulus in this case acts as a prime polynomial which means that no other polynomial can divide this polynomial.

Such polynomial are refer to as irreducible polynomials or prime polynomials.

Eg:

Degree	irreducible polynomial
1	$x+1, x, x-1$
2	x^2+x+1
3	x^3+x^2+1, x^3+x+1
4	$x^4+x^3+x^2+x+1$, x^4+x^3+1, x^4+x+1
5	$x^5+x^2+1, x^5+x^3+x^2+x+1$, etc

Q Add the following polynomials in GF(2^8)

→ $x^5+x^2+x, x^7+x^4+x^3+x, x^4-x^3+x+1$

⇒ $x^7+x^5+x^4+x^4+x^3-x^3+x^2+x^2+x+x+x+1$

⇒ $x^7+x^5+(1+1)x^4+(1-1)x^3+(1+1)x^2+(1+1+1)x^2+1$

⇒ $x^7+x^5+0x^4+0x^3+0x^2+1x+1$

$\Rightarrow x^7 + x^5 + x + 1$

Q. Multiply following polynomials in $GF(2^8)$ with an irreducible polynomial.

$\rightarrow x^8 + x^4 + x^3 + x + 1$.

$(x^5 + x^2 + x) \times (x^7 + x^4 + x^3 + x^2 + x)$

$\Rightarrow x^5(x^7 + x^4 + x^3 + x^2 + x) + x^2(x^7 + x^4 + x^3 + x^2 + x) + x(x^7 + x^4 + x^3 + x^2 + x)$

$= x^{12} + x^9 + x^8 + x^7 + x^6 + x^9 + x^6 + x^5 + x^4 + x^3 + x^8 + x^5 + x^4 + x^3 + x^2$

$\Rightarrow x^{12} + x^9 + x^9 + x^8 + x^8 + x^7 + x^6 + x^6 + x^5 + x^5 + x^4 + x^4 + x^3 + x^3 + x^2$

~~$\Rightarrow x^{12} + (1+1)x^9 + (1+1)x^8 + x^7 + (1+1)x^6 + (1+1)x^5 + (1+1)x^4 + (1+1)x^3 + x^2$~~

$\Rightarrow x^{12} + 0x^9 + 0x^8 + x^7 + 0x^6 + 0x^5 + 0x^4 + 0x^3 + x^2$

$\Rightarrow (x^{12} + x^7 + x^2) \bmod x^8 + x^4 + x^3 + x + 1$

$x^8 + x^4 + x^3 + x + 1 \;\big|\; x^{12} + x^7 + x^2 \;\big|\; x^4 + x + 1$

$x^{12} + x^8 + x^7 + x^5 + x^4$

$\oplus$

$x^8 + x^5 + x^4 + x^2$

$x^8 + x^4 + x^3 + x + 1$

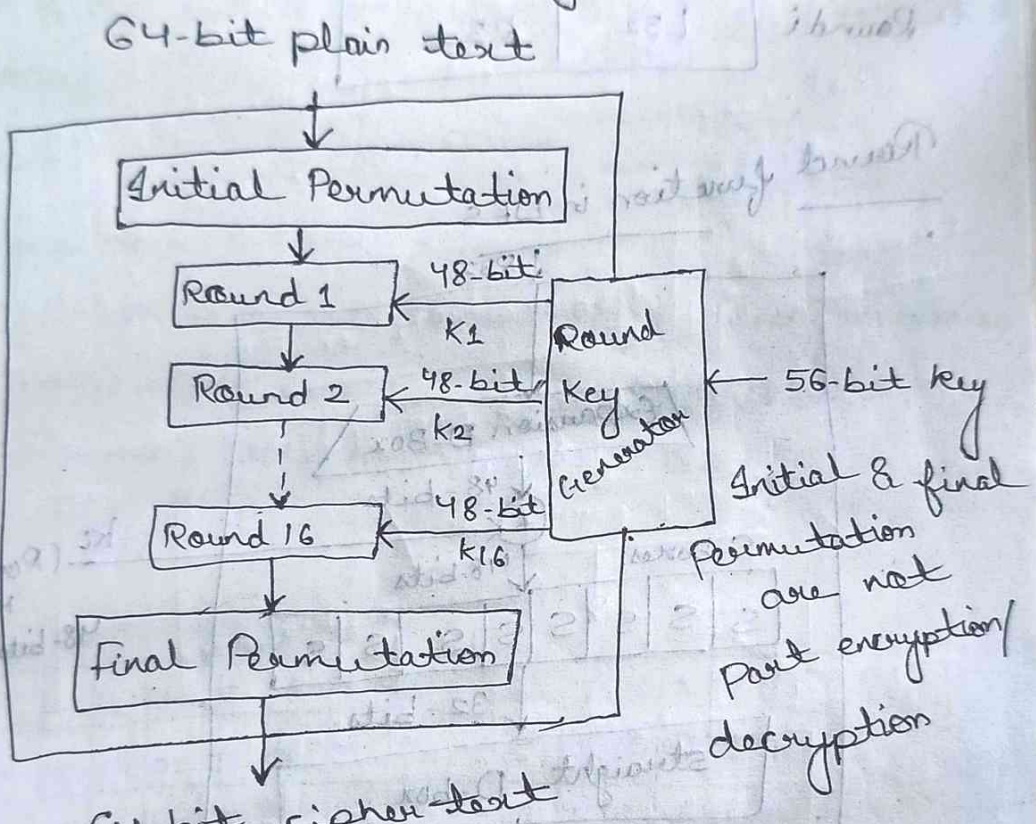
$x^5 + x^3 + x^2 + x + 1$

$= \boxed{x^5 + x^3 + x^2 + x + 1}$

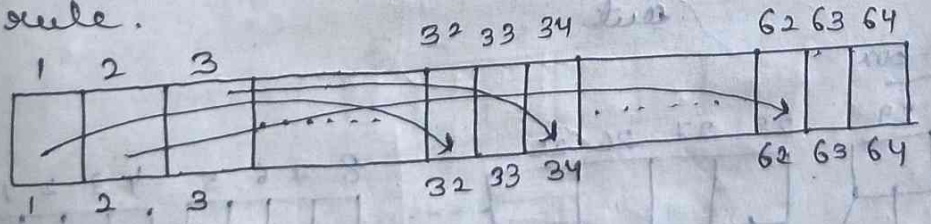
20/08/26 Modern symmetric key cryptography

Data Encryption Standard (DES)

- It is a block cipher, published by NIST, USA.
- It is a block cipher that takes a block of 64-bits as input & produces a cipher text of 64-bit.
- It uses a 56-bit key for encryption & decryption.
- DES uses 16 rounds for encryption.

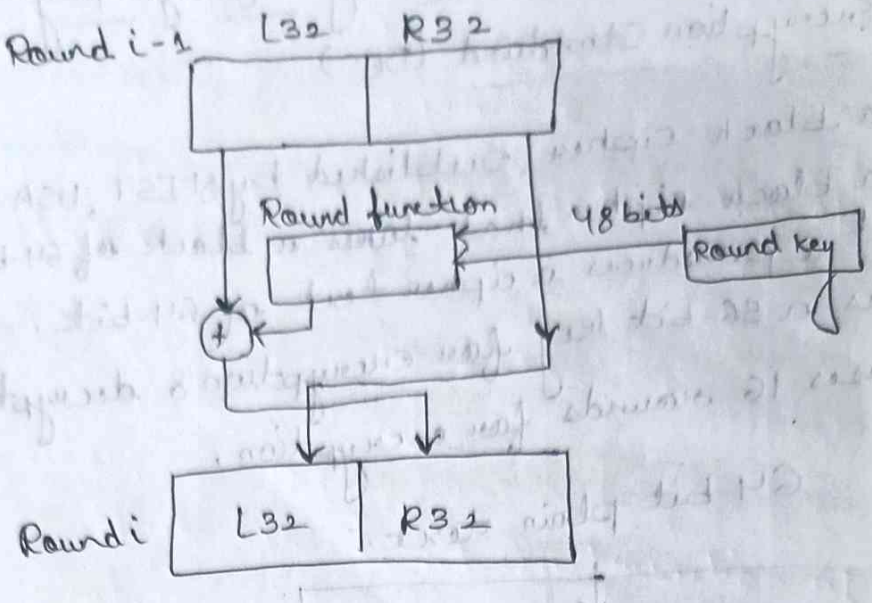


- Intial permutation is based on a predefined rule.

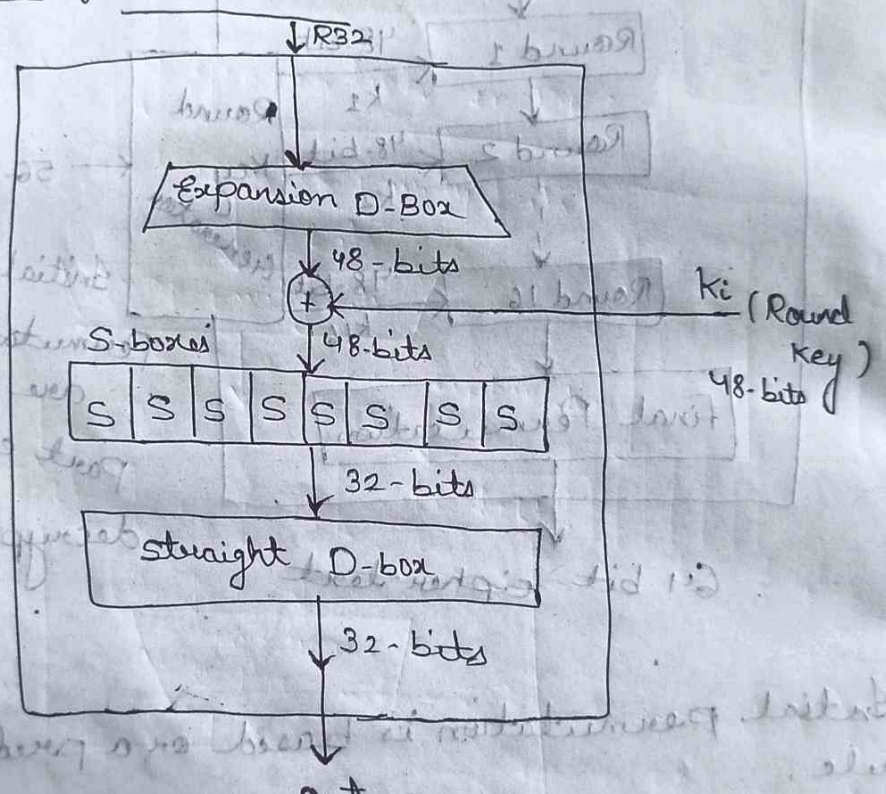


- Initial & final permutation are reverse of each other.

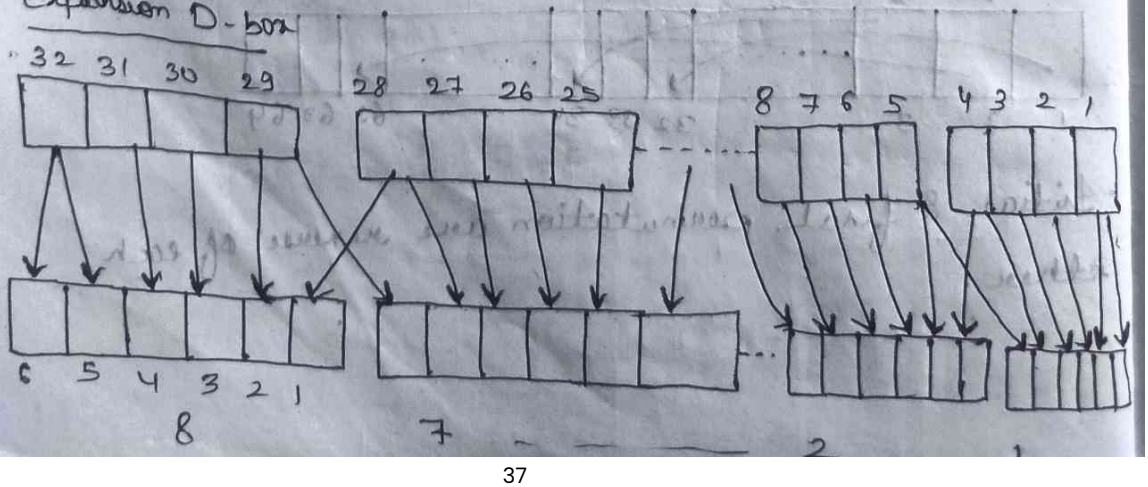
Round of DES



Round function in DES



Expansion D-box



→ The 32-bit input is divided into 8 4-bits sections

→ Each 4-bit section is then expanded to 6-bits by using a pre-defined rule.

→ For each section input bits 1, 2, 3 & 4 are copied to output bits 2, 3, 4 & 5.

→ Output bit 1 comes from bit 4 of the previous section & o/p bit 6 comes from bit 1 of the next section.

21/08/26 **Substitution S-box**

→ There are 8-boxes

→ Every S-box is a 2-D table (4×16) having 64 entries

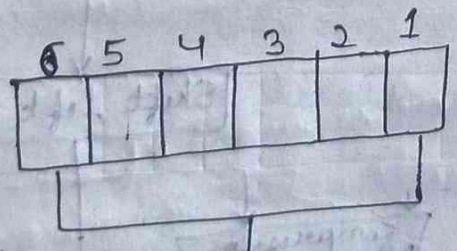
→ The rows are numbered as 0, 1, 2, 3 (00, 01, 10, 11)

→ The columns are numbered as 0 to 15 (00 01 — — A, B, C, D, E, F)

S-box 1

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	00	09	07
1	00	15	07	04	14	02	13	10	03	06	12	11	09	03	05	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	05	10	00
3	15	12	08	02	04	04	01	07	05	11	03	14	10	06	10	13

Input 6-bits



Row no.

Combination of 2, 3, 4, 5 serves as the column no.

Eg:- Input : 101101

Row no. = 11 = 3

column no. = 0110 = 6

The value obtained at intersection of row 3 & col. 6 is 01 (0001)

Input : 111001

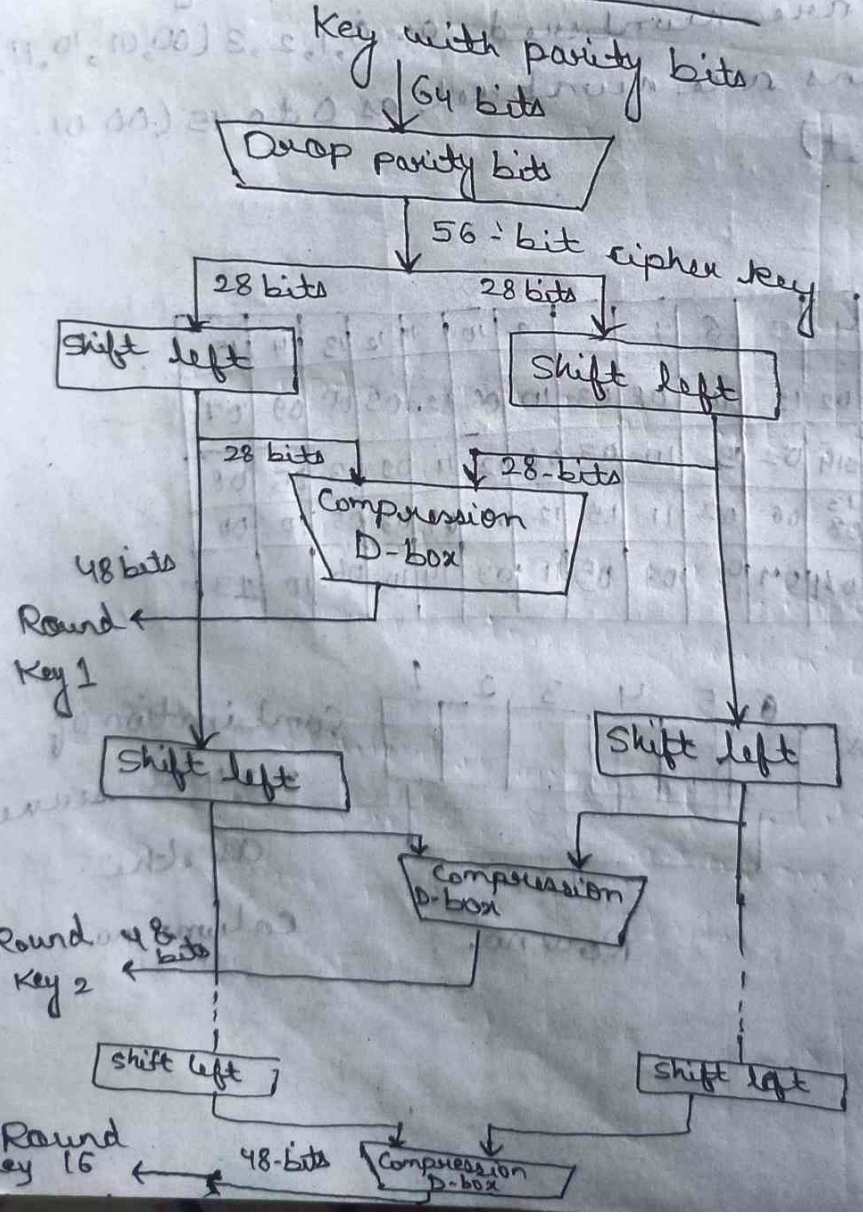
Row no : = 11 = 3

Col. no. = 1100 = 12

output = value at intersection of row 3 & col. 12.

= 1010

Round - key generation in DES



→ No. of places for shifting of the bits depends on the round no.

Rounds	Shift
1, 2, 9, 16	one bit
others	two bits

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Analysis of strength of DES

→ DES satisfies two desired properties of block ciphers.

1) Avalanche Effect

→ It means a small change in the plain text or in the key should create a significant change in the cipher text.

→ The DES has been proved to be strong with regard to this property.

ii) Completeness Effect

→ Completeness effect means that each bit of the cipher text needs to depend on many bits of the plain text.

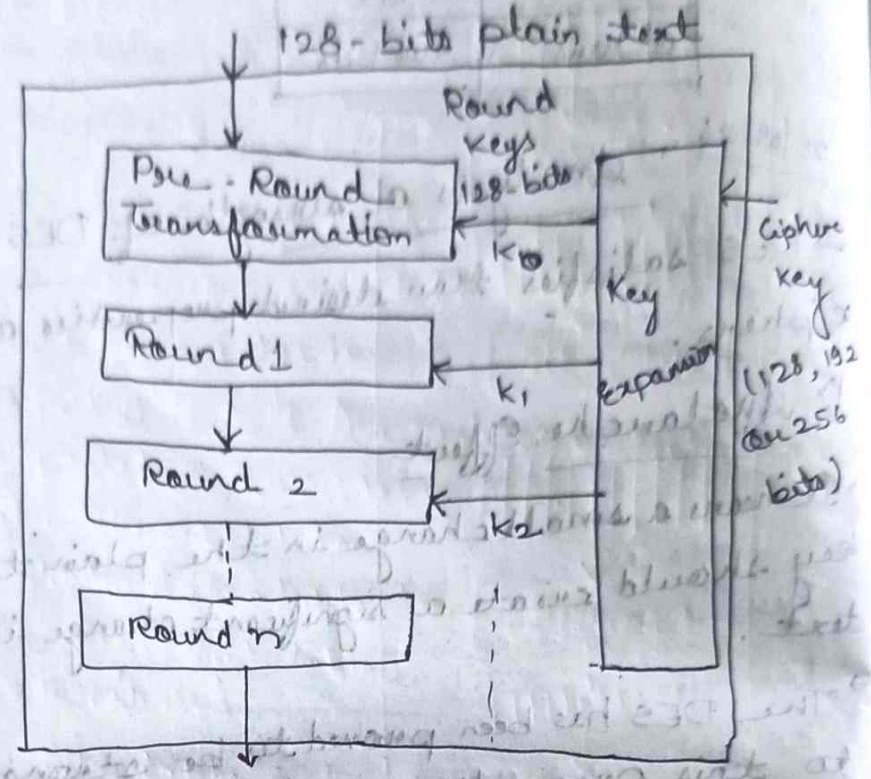
→ The diffusion & confussion produced by D-boxes & S-boxes shows a very strong completeness effect.

→ In practise double DS equals to triple DS - 3DS are used with key length of 112 bits & 168 bits respectively.

Advanced encryption standard (AES)

→ The AES is a symmetric Key block cipher published by National Institution of standard & Technology (NIST).

→ It uses a block of plain text of 128 bits & three different key sizes of 128, 192 & 256 bits.



n	Key size
10	128
12	192
14	256

→ AES uses data / block that can be of 5 types

i) bit - either 1 or 0

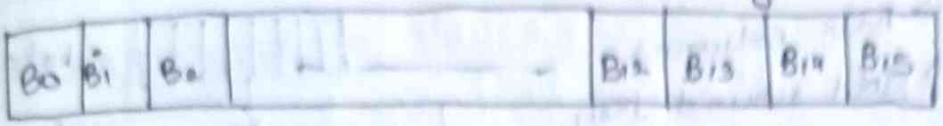
ii) byte - combination of 8 bits. Represented as either a row matrix (1×8) or a column matrix (8×1)

iii) word - A word is group of 32 bits that can be created as a single entry or a row matrix of 4 bytes or a column matrix of 4 bytes.

word (32 bits) → $\begin{bmatrix} B_0 & B_1 & B_2 & B_3 \end{bmatrix} \rightarrow \begin{bmatrix} B_0 \\ B_1 \\ B_2 \\ B_3 \end{bmatrix}$

iv) Block

A block in AES is a group of 128 bits & it can be represented as a state matrix of 16 bytes.



Block

v) State

AES uses several rounds in which each round is made of several stages.

Data block is transformed from one stage to another.

The data block before & after every stage is referred to as a state (S).

A state is normally treated as a 4×4 matrix.

$$\begin{bmatrix} b_0 & b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 & b_7 \\ b_8 & b_9 & b_{10} & b_{11} \\ b_{12} & b_{13} & b_{14} & b_{15} \end{bmatrix}$$

01/09/26 Advance encryption standard (AES)

$S_i \bmod 4, i/4 \leftarrow b_i$

$S_0 \bmod 4, 0/4 \leftarrow b_0 \rightarrow b_0$ stored at $S_{0,0}$

$b_1 \rightarrow S_{1,0}$

$b_2 \rightarrow S_{2,0}$

$b_3 \rightarrow S_{3,0}$

→ A block is converted to state by their formula.

→ A state is converted to block by using

$$b_{i+4j} \leftarrow S_{i,j}$$

Eg:- $S_{2,1} \rightarrow b_{2+4.1} = b_6$

$S_{3,2} \rightarrow b_{3+4.2} = b_{11}$

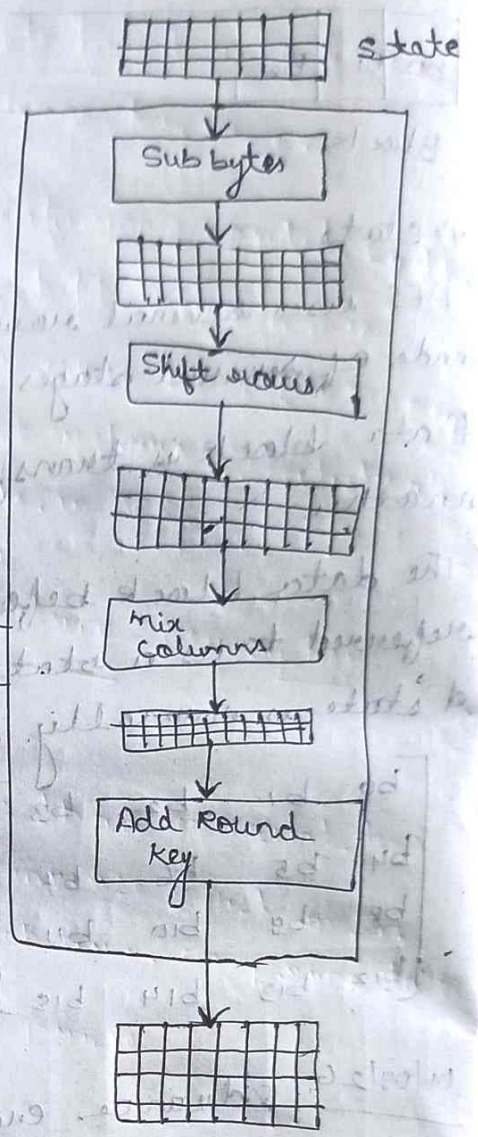
→ Every round in AES undergoes 4 stages

i) Sub bytes (substitute bytes)

ii) Shift Rows

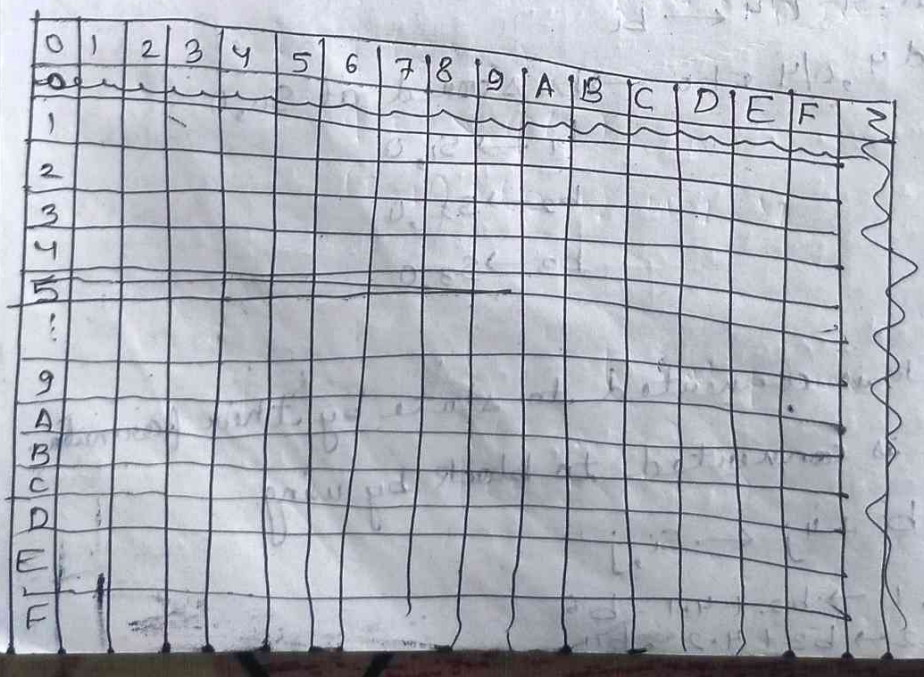
iii) Mix columns

iv) Add round key



i) Sub bytes

Transformation sub bytes is used at the encryption side. A byte is interpreted as 2-hexadecimal. The substitution of byte is preformed by a table look-up the resultant table is of 16 rows & 16 columns. Every entry in the table is also a byte interpreted as 2-hexadecimal digit.



→ The first 4 bits of i/p byte serve as a row no. & next 4 bits serve as a column no.

→ The i/p byte is substituted as values in the table are the intersection of row & column.

ii) Shift Rows

It is used as the permutation of bytes in this operation every byte is shifted to left.

The no. of shifts depends upon the row no.

Row 0 = no shift

Row 1 = Shift left by 1

Row 2 = Shift left by 2

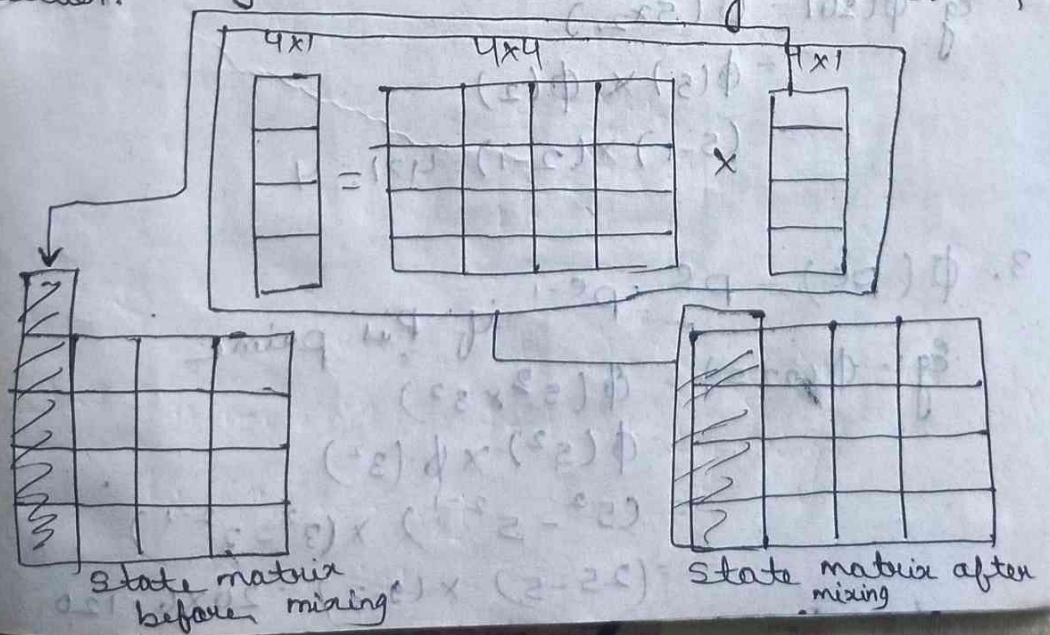
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iii) Mix columns

The mix column operation changes the contents of each byte by taking 4 bytes at a time & combining them to recreate 4 new bytes.

→ To ensure the difference between the bytes before & after mixing, each input byte is multiplied with a different constant.

→ The mixing can be provided by matrix multiplication.



Strength of AES

→ AES is more secure than DES due to the larger key size.

→ This is almost impossible for brute force attack in AES due to large key size.

→ The strong diffusion & confusion provided by the combination of sub bytes, shift rows & mix column transformation removes any frequency pattern in the plain text. Therefore, statistical attack is also not possible in AES.

10/09/25

Mathematics of assymetric key ciphers

Euler's phi function

$\phi(n)$ = no. of integers less than or equal to n whose multiplicative inverse exist in Z_n (i.e no. of integers which are co-prime to n).

1. $\phi(n) = n-1$ if n is prime

Eg: $\phi(13) = 12$

2. $\phi(m \times n) = \phi(m) \times \phi(n)$ if m & n are coprime

Eg. $\phi(10) = \phi(5 \times 2)$

$= \phi(5) \times \phi(2)$

$= (5-1) \times (2-1) = 4 \times 1 = 4$

3. $\phi(p^e) = p^e - p^{e-1}$ if p is prime

Eg: $\phi(225) = \phi(5^2 \times 3^2)$

$= \phi(5^2) \times \phi(3^2)$

$= (5^2 - 5^{2-1}) \times (3^2 - 3^{2-1})$

$= (25-5) \times (9-3) = 20 \times 6 = 120$

4. $\phi(1) = 0$

5 | 240
3 | 48
2 | 16
2 | 8
2 | 4
2

(a) $\phi(240)$

$= \phi(16 \times 3 \times 5)$

$= \phi(16) \times \phi(3) \times \phi(5)$

$= \phi(2^4) \times \phi(3) \times \phi(5)$

$= (2^4 - 2^{4-1}) \times (3-1) \times (5-1)$

$= (16-8) \times 2 \times 4 = 8 \times 2 \times 4 = 64$

(b) $\phi(80)$

$= \phi(16 \times 5)$

$= \phi(16) \times \phi(5)$

$= \phi(2^4) \times \phi(5-1)$

$= (2^4 - 2^3) \times 4$

$= (16-8) \times 4$

$= 8 \times 4 = 32$

(c) $\phi(100)$

$= \phi(5^2 \times 2^2)$

$= \phi(5^2) \times \phi(2^2)$

$= (5^2 - 5^{2-1}) \times (2^2 - 2^{2-1})$

$= (25-5) \times (4 \times 2)$

$= 20 \times 2$

$= 40$

Fermat's little theorem

First version :- It states that if p is a prime no. & a is an integer such that p does not divide a, then $a^{p-1} \equiv 1 \bmod p$

Eg :- $6^{10} \bmod 11 = 6^{11-1} \bmod 11 = 1 \bmod 11$

Eg: $5^{12} \bmod 13$

$= 5^{13-1} \bmod 13$

$= 1 \bmod 13$

Eg: $5^{13} \bmod 13$

$= (5^{12} \times 5) \bmod 13$

$= (5^{12} \bmod 13 \times 5 \bmod 13) \bmod 13$

$= (1 \times 5) \bmod 13$

$= 5 \bmod 13$

$= 5^{13-1} \bmod 13$

$= 1 \bmod 13$

Eg:- $21^6 \equiv 1 \bmod 7$

(FLT first version doesn't work)

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Second Version

It states that if P is a prime & a is an integer then $\boxed{a^p \equiv a \bmod p}$

Eg:- $8^{\overset{a}{7}} \bmod 7$

$= 8 \bmod 7$

$= 1 \bmod 7$

$= (8 \bmod 7)^7 \bmod 7$

$= 1^7 \bmod 7$

Eg:- $15^{23} \bmod 23$

$= (15^2)^{11} \times 15 \bmod 23$

$= (225 \bmod 23)^{11} \bmod 23 \times 15 \bmod 23$

$= (18)^{11} \bmod 23 \times 15 \bmod 23$

$= (18^2)^5 \times 18 \bmod 23 \times 15 \bmod 23$

$= 324^5 \times 18 \bmod 23 \times 15 \bmod 23$

$= 2^5 \bmod 23 \times 18 \bmod 23 \times 15 \bmod 23$

$= 9 \bmod 23 \times 18 \bmod 23 \times 15 \bmod 23$

$= (9 \times 18) \bmod 23 \times 15 \bmod 23$

$= 162 \bmod 23 \times 15 \bmod 23$

$= 1 \times 15 \bmod 23$

$= 15 \bmod 23$

Eg:- $65^{101} \bmod 101$

$= 65 \bmod 101$

23 | 324 (14
23
94
92
2

Eg:- $65^{102} \bmod 101$

$= 65^{101} \times 65 \bmod 101$

$= (65^{101} \bmod 101 \times 65 \bmod 101) \bmod 101$

$= (65 \times 65) \bmod 101$

$= 4225 \bmod 101$

$= 84 \bmod 101$

→ Fermat little theorem can be used to find multiplicative inverse of an integer modulo prime no. without using extended euclidian algorithm.

$a^{p-1} \equiv 1 \bmod p$

$a \times a^{-1} \bmod p = 1 \bmod p$

$a \times a^{-1} \bmod p = a^{-1} \times a^{p-1} \bmod p$

$= a^{p-2} \bmod p$

Eg:- $8^{-1} \bmod 17$

$= 8^{17-2} \bmod 17$

$= 8^{15} \bmod 7$

$= (8^3)^5 \bmod 17$

$= 2^5 \bmod 17$

$= 32 \bmod 17$

$= 15 \bmod 17$

Eg:- $7^{-1} \bmod 19$

$= 7^{19-2} \bmod 19$

$= 7^{17} \bmod 19$

$= (7^2)^8 \times 7 \bmod 19$

$= (7^2 \bmod 19)^8 \times 7 \bmod 19$

$= (11)^8 \bmod 19 \times 7 \bmod 19$

$= (11^2 \bmod 19)^4 \times 7 \bmod 19$

$= (7)^4 \bmod 19 \times 7 \bmod 19$

Eg:- $5^{-1} \bmod 23$

$5^{23-2} \bmod 23$

$= 5^{21} \bmod 23$

$= (5^{-2})^{10} \times 5 \bmod 23$

$= (5^2 \bmod 23)^{10} \times 5 \bmod 23$

$= 2^{10} \bmod 23 \times 5 \bmod 23$

$= (2^5 \bmod 23)^2 \times 5 \bmod 23$

$= (9 \bmod 23)^2 \times 5 \bmod 23$

$= 12 \times 5 \bmod 23$

$= 60 \bmod 23$

$= 14 \bmod 23$

$= 121 \bmod 19 \times 7 \bmod 19$

$= 7 \times 7 \bmod 19$

$= 49 \bmod 19$

$= 11 \bmod 19$

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Euler's theorem

First version

→ The first version of Euler's theorem states that if two integers a & n are co-prime

$$a^{\phi(n)} \equiv 1 \bmod n$$

Eg:- $5^6 \bmod 7$

$= 5^{\phi(7)} \bmod 7 = 1 \bmod 7$

$5^6 \bmod 7$

$(5^2)^3 \bmod 7 = (5^2 \bmod 7)^3 \bmod 7$

$= 4^3 \bmod 7 = 4^2 \times 4 \bmod 7$

$= ((16 \bmod 7 \times 4 \bmod 7) \bmod 7$

$= 1 \bmod 7$

Second version

→ The 2nd version states that if a & n are any integers such that

$n = p \times q$, $a < n$ & k is an integer then

$$a^{k \times \phi(n) + 1} \equiv a \bmod n$$

$a = 5 \quad n = 42$

$\phi(42) = \phi(2) \times \phi(3) \times \phi(7)$

$= 2 \times 6 = 12$

$5^{25} \bmod 42$

$= 5^{2 \times 12 + 1} \bmod 42$

$= 5^{2 \times \phi(42)+1} \bmod 42$

$= (5^3)^8 \times 5 \bmod 42$

$= (125 \bmod 42)^8 \times 5 \bmod 42$

$= 41^8 \bmod 42 \times 5 \bmod 42$

$= (41^2)^4 \bmod 42 \times 5 \bmod 42$

$= (1681)^4 \bmod 42 \times 5 \bmod 42$

$= 1 \bmod 42 \times 5 \bmod 42$

$= 5 \bmod 42$

$= 1 \times 2 \times 6$

$= 12$

$= 5 \bmod 42$

Chinese Remainder Theorem

It is used to solve a system of congruent equations with one variable for different moduli.

$x \equiv a_1 \bmod m_1$

$x \equiv a_2 \bmod m_2$

$\vdots$

$x = a_n \bmod m_n$

Steps to solve a set of linear congruent equation by using chinese remainder theorem

Step1 - find $M = m_1 \times m_2 \times \dots \times m_n$

Step2 - find $M_1 = M/m_1$, $M_2 = M/m_2$ ...

Step3 - find the multiplicative inverse of M_1, M_2 ... w.r.t the moduli m_1, m_2 ...

Step 4 - The solution to the simultaneous equation is $x = (a_1 \times M_1 \times M_1^{-1} + a_2 \times M_2 \times M_2^{-1} \ldots) \bmod M$

Q Solve $x \equiv 2 \bmod 3$

$x \equiv 3 \bmod 5$

$x \equiv 2 \bmod 7$

$M = 3 \times 5 \times 7 = 105$

$M_1 = M/m_1 = \frac{105}{3} = 35$

$M_2 = M/m_2 = \frac{105}{5} = 21$

$M_3 = M/m_3 = \frac{105}{7} = 15$

$M_1^{-1} = 35^{-1} \bmod 3$

$= 2$

$M_2^{-1} = 21^{-1} \bmod 5$

$= 1 \bmod 5$

$M_3^{-1} = 15^{-1} \bmod 7$

$= 1$

$x = (2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1) \bmod 105$

$\equiv 233 \bmod 105$

$= 23 \bmod 105$

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$x \equiv 3 \bmod 7$

$x \equiv 5 \bmod 9$

$x \equiv 3 \bmod 14$

$M = 7 \times 9 \times 14 = 882$

$M_1 = \frac{M}{m_1} = \frac{882}{7} = 126$

$M_2 = \frac{M}{m_2} = 98$

$M_3 = \frac{M}{m_3} = 63$

$M_1^{-1} = 126^{-1} \bmod 7$

No soln exist

Q $x \equiv 7 \bmod 13$

$x \equiv 11 \bmod 12$

$M = 13 \times 12 = 156$

$M_1 = \frac{M}{m_1} = \frac{156}{13} = 12$

$M_2 = \frac{M}{m_2} = \frac{13 \times 12}{12} = 13$

$M_1^{-1} \bmod 13 = 12^{-1} \bmod 13 = 12 \bmod 13$

$M_2^{-1} \cdot M \bmod 12 = 13^{-1} \bmod 12 = 1 \bmod 12$

$x = (a_1 \times M_1 \times M_1^{-1} + a_2 \times M_2 \times M_2^{-1})$

$= (7 \times 12 \times 12 + 11 \times 13 \times 1)$

$= 1151 \bmod 156$

$= 59 \bmod 156.$

Quadratic Congruence

The relation of the form $x^2 \equiv a \bmod n$ is known as a quadratic congruence.

The sol^n is $x \equiv \pm\sqrt{a} \bmod n$

→ The relation $x^2 \equiv a \bmod n$ is said to be a quadratic residue (QR) in mod n if it has two sol^{n}s.

→ It is said to be a quadratic non-residue (QNR) if it doesn't have any sol^n.

The quadratic congruence modulo a prime no.

Eg: $Z_7^* = \{1, 2, 3, 4, 5, 6\}$

$x^2 \equiv 1 \bmod 7$

$x^2 \equiv 2 \bmod 7$

$x^2 \equiv 3 \bmod 7$ (QNR)

$x^2 \equiv 4 \bmod 7$ ✓

$x^2 \equiv 5 \bmod 7$ (QNR)

$x^2 \equiv 6 \bmod 7$ (QNR)

↑ quadratic congruent relation $(x^2 \equiv a \bmod p)$ modulo a prime no "p" is said to be a QR ~~if~~ in Z_p if $\boxed{a^{\frac{p-1}{2}} \equiv 1 \bmod p}$

→ It is QNR in Z_p if $\boxed{a^{\frac{p-1}{2}} \equiv -1 \bmod p}$

Eg: $x^2 \equiv 5 \bmod 7$ QNR

$a^{\frac{p-1}{2}} \equiv -1 \bmod p$

$5^{\frac{7-1}{2}} \equiv -1 \bmod 7$

$5^3 = 125 \bmod 7$

~~$= 17 \bmod$~~ $= 6 \bmod 7$

$= -1 \bmod 7$

7 | 225 | 1
7
55
49
6

Q $x^2 \equiv 14 \bmod 23$ is a QR or QNR

$= 14^{\frac{23-1}{2}} \bmod 23$

$= 14^{11} \bmod 23$

$= (14^2)^5 \times 14 \bmod 23$

$= (196)^5 \times 14 \bmod 23$

$= 12^5 \bmod 23 \times 14 \bmod 23$

$= (12^2)^2 \times 12 \bmod 23 \times 14 \bmod 23$

$= (144)^2 \times 12 \bmod 23 \times 14 \bmod 23$

$= (6)^2 \times 12 \bmod 23 \times 14 \bmod 23$

$= (13 \times 12) \bmod 23 \times 14 \bmod 23$

$= 156 \bmod 23 \times 14 \bmod 23$

$= 18 \bmod 23 \times 14 \bmod 23$

$= 18 \times 14 \bmod 23$

$= -1 \bmod 23$

23 | 252 | 10

23

22

if remainder n-1 then equivalent to -1

Solving a quadratic congruence modulo of a prime no.

Prime numbers can be expressed as $4K+1$ or $4K+3$ (where K is a the integer)

$x^2 \equiv a \bmod p$

$x \equiv \pm \sqrt{a} \bmod p$

$\sqrt{a} = a^{\frac{p+1}{4}} \pmod p$

$7^{\frac{19-1}{2}} \bmod 19 = 1 \bmod 19$

$x^2 \equiv 7 \bmod 19$

$7^{\frac{19-1}{2}} \bmod 19 \equiv \bmod 19$

$7^9 \bmod 19$

$(7^2)^4 \times 7 \bmod 19$

$11^4 \times 7 \mod 19$

$(7 \times 7 \times 7) \mod 19$

$= 1 \mod 19$

$19 \overline{)121} \; (6$
$\underline{114}$
7

$x_1 = 7^{\frac{19+1}{4}} \mod 19$

$= 7^5 \mod 19$

$= 7^3 \times 7^2 \mod 19$

$= 49 \mod 19$

$= 11 \mod 19$

$x_2 = -7^{\frac{19+1}{4}} \mod 19$

$x_2 = -11 \mod 19$

$= 8 \mod 19$

$x_1 = 11 \mod 19$

$x_2 = 8 \mod 19$

(6) $x^2 \equiv 2 \mod 11$

$2^{\frac{11+1}{2}} \mod 11$

$2^5 \mod 11$

$(2^3)3 \mod 11$

$32 \mod 11$

$= -1 \mod 11$

06/09/25

* Asymmetric key Cryptography / public key encryption

RSA cryptosystem

↓

Rivest Shamir, Adleman

→ RSA cryptosystem uses modular exponentiation for encryption & decryption.

→ It uses the algebraic structure ring for encryption & decryption & a group for key generation.

Public ring $R = \langle \mathbb{Z}_n, +, \times \rangle$

Private group $G = \langle \mathbb{Z}^*_{\phi(n)}, \times \rangle$

RSA Algorithm for key generation

1) Select two large prime numbers p & q such that $p \neq q$

2) $n = p \times q$

3) $\phi(n) = (p-1) \times (q-1)$

4) Select an integer e such that $1 < e < \phi(n)$

$\gcd(e, \phi(n)) = 1$

5) $d = e^{-1} \bmod \phi(n)$

6) Public key = (e, n)

7) Private key = (d, n)

Encryption

$$C = P^e \bmod n$$

Decryption

$$P = C^d \bmod n$$

Eg:- $P = 7, q = 11$

$n = P \times q = 7 \times 11 = 77$

$\phi(n) = \phi(77) = 6 \times 10 = 77$

$\downarrow$

$(7-1) \times (11-1) = (6 \times 10)$

$e = 13$

Public key = $(13, 77)$

$d = e^{-1} \bmod 60 = 13^{-1} \bmod 60$

$= 37$

$37 \times 13 = 481$

$60 \overline{)481}\ 8$

480

1

Private key = $(37, 77)$

P = 'F' = 5

$C = 5^{19} \bmod 77$

$C = (5^3)^4 \times 5 \bmod 77$

$= (48^4 \bmod 77) \times 5 \bmod 77$

$= 71^2 \bmod 77 \times 5 \bmod 77$

$= 36 \times 5 \bmod 77$

$= 26 \bmod 77$

09/10/26

Strength of RSA

The possible attacks on RSA can be :-

i) Factorization attack

ii) Encryption exponent

iii) Decryption exponent

iv) Modulus attack

v) Implementation attack

No drastic attack on RSA have been yet discovered. However the above attacks have been predicted based on the weak plain text, weak parameter selection or inappropriate implementation.

Elgamal crypto system

It uses the concept of primitive root & discrete logarithm.

Key generation

→ Select a large prime no. p

→ Select an integer d to be a member of the group $G_1 = (Z_p^*, X)$

→ Select e_1 be a primitive root in the group $G_1 = (Z_p^*, X)$

→ ~~Select e_2~~ Calculate $e_2 = e_1^d \bmod p$

→ Public key = ~~(e1, e2, p)~~ (e_1, e_2, p)

→ Private key = (d, p)

Encryption

→ Select a random integer γ in the group $G_1 = (Z_p^*, X)$

→ $c_1 = e_1^{\gamma} \bmod p$

$c_2 = (P \times e_2^{\gamma}) \bmod p$

Decryption

$$P = \left[c_2 (c_1^d)^{-1} \right] \bmod p$$

$Z_5^* = \{1, 2, 3, 4\}$

Z_p^* =

The order of an element $x < \phi(p)$ in Z_p^* is the smallest integer i, such that $x^i \bmod p \equiv 1 \bmod p$.

if $i = \phi(p)$, then x is said to be a primitive root in Z_p^*

$Z_5^* = \{1, 2, 3, 4\}$ $\quad \phi(5) = 5 - 1 = 4$

$1^1 \bmod 5 \equiv 1 \bmod 5$

$2^1 \bmod 5 \equiv 2 \bmod 5$, $2^2 \bmod 5 \equiv 4 \bmod 5$

$2^3 \bmod 5 = 3 \bmod 5$, $2^4 \bmod 5 = 1 \bmod 5$

Order of $2 = 4$, 2 is a primitive root bcz order of $2 = 4$ & $\phi(5) = 4$

~~$3^1 \bmod 5 = 3$, $3^2 \bmod$~~

$3^1 \bmod 5 = 3 \bmod 5$, $3^2 \bmod 5 = 4 \bmod 5$

$3^3 \bmod 5 = 2 \bmod 5$, $3^4 \bmod 5 = 1 \bmod 5$

$4^1 \bmod 5 = 4 \bmod 5$, $4^2 \bmod 5 = 1$

Order of $4 = 2 \neq \phi(5)$ so 4 is not a primitive root only 2 & 3 are primitive root.

Q Whether 3 is a primitive root in Z_7^*

Sol:- $Z_7^* = \{1, 2, 3, 4, 5, 6\}$ $\phi(7) = 7-1 = 6$

~~$1 \bmod 7 = 1 \bmod 7$~~

$3^1 \bmod 7 = 3 \bmod 7$, $3^2 \bmod 7 = 2 \bmod 7$,

$3^3 \bmod 7 = 6 \bmod 7$, $3^4 \bmod 7 = 4 \bmod 7$,

$3^5 \bmod 7 = 5 \bmod 7$, $3^6 \bmod 7 = 1 \bmod 7$

Order of 3 = 6 = $\phi(7)$

So 3 is a primitive root.

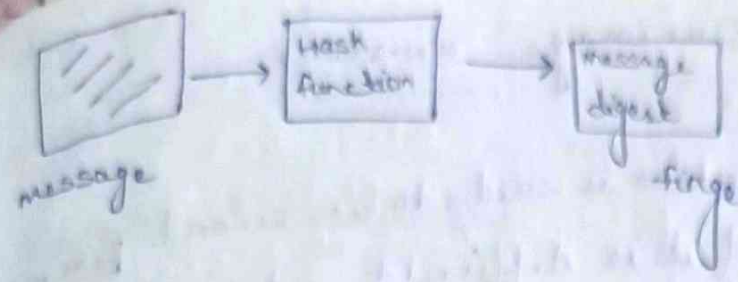
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Message Integrity & Authentication

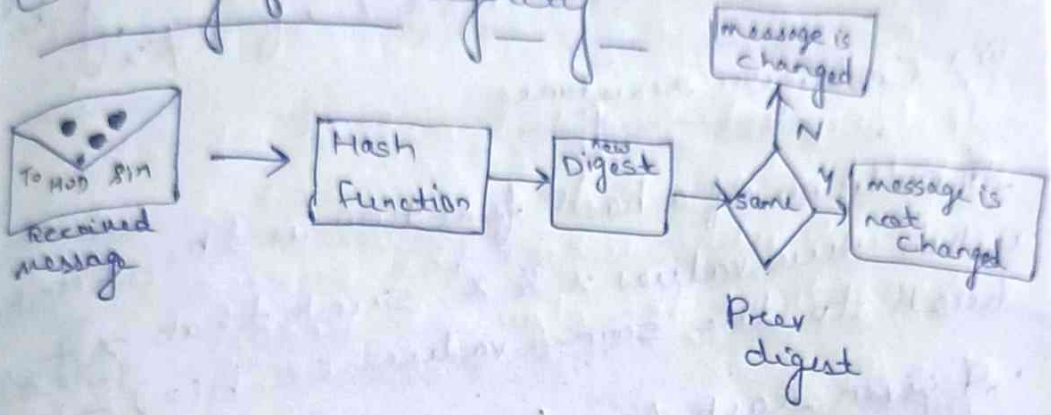
Message digest & Digital signature.

Message digest

- A message digest is the electronic equivalent of the document & fingerprint.
- It is used to preserve the integrity of a message.
- In order to preserve the integrity the message is subjected to a cryptography hash function.
- The Hash function is used to create a compressed image of the original message.



Checking for integrity



Properties of a cryptographic hash function

→ It must have 3 properties :-

1) Pre image resistance

It implies that ~~giving~~ given a hashed value it should be difficult for an ~~adversary~~ adversary to compute the pre image of the hashed value.

A hash function h is said to be pre-image resistant if, given the value $y = h(x)$, for some x, it is difficult to compute x' such that $h(x') = y$

ii) Second preimage resistance

. A hash function is said to be second preimage resistance if it is difficult adversary to compute $h(x) = h(x')$ if $x \neq x'$.

iii) Collision resistance

A collision of a hash function is the event when two values x & x' such that $x \neq x'$ hash to the same value i.e $h(x) = h(x')$.

. A given hash function is said to be collision resistance if it is difficult for an adversary to find the collision. for two different values x & x'.

Randome oracle model

A hash function based on this model behaves as follows :-

i) When a new message of any length is given the oracle creates & gives a fix length message digest which is a random strings of 0's & 1's. The oracle records the message & digest.

ii) When a message is given for which a digest exist, the oracle simply gives the digest in the records.

iii) The digest for a new message needs to be choosen independently from all previous

digest.

04/11/20

Entity Authentication

Modification detection code (MDC)
Message authentication code (MAC)

An entity can be a client, server, Person, machine etc.

Data - origin verification / message authentication.
Entity Authentication.

An entity authentication is a technique designed to let one party ~~of~~ to prove identity of another party.

- The entity whose identity needs to be proved is called the claimant.
- The entity that ~~tries~~ tries to prove the identity of claimant is called the verifier.

Message Authentication	Entity Authentication
→ The receiver & sender may not be online. Eg:- mail message	→ The receiver & sender must be online. Eg:- online chating
→ Message authentication simply authenticates one message at a time so that the process need to be	

updated for each message.

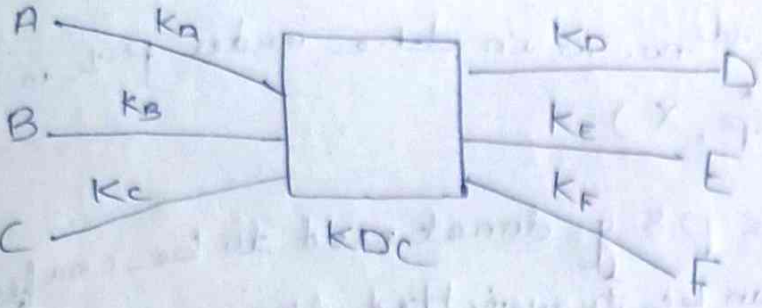
- Entity authentication authenticates the claimant for the entire duration of a session.

- The simplest & oldest method of entity authentication is the password based authentication where the password is something that the claimant knows.
- A password is used when a user needs to access a system to use the system resources.
- Each user as a user identification i.e public & a password i.e private.
- There are 2 types of password used for entity authentications are :-
  i) fixed password
  ii) One time password (OTP)

Key Management

Key distribution center (KDC)

- A KDC is a trusted third party used for managing keys (symmetric key distribution)
- A secret key is established by each entity & is shared with the KDC.



When an entity A wants to send a confidential message to entity D.

i) A sends the request to KDC stating that it needs a session secret key between A & D.

ii) KDC informs the entity D for edge request

iii) If D agrees, a session key is created between the two.

iv) The secret key b/w A & D that is established with KDC is used to authenticate A & D to the KDC.

05/11/26

Diffie-Hellman Key Exchange

- According to Diffie Hellman protocol two Parties create a symmetric session key without the need of KDC.
- Before establishing a symmetric key, the two Parties need to choose two large prime nos. "P & q". These prime nos are on the order of 300 decimal digits (1024 bits).

- The 2nd no. is in the order p-1 in the group $(Z_p^*, \times)$
- Two nos p & g donot need to be confidential & they can be transmitted through insecured channel.

Diffie Hellman Key Exchange

Steps

i) The sender entity 'A' chooses a large random no. 'x' such that $0 \leq x \leq p$. or $0 \leq x < p-1$, & calculates $R_1 = g^x \bmod p$.

ii) The receiver entity 'B' chooses another large random no. y such that $0 < y \leq p-1$ & calculates $R_2 = g^y \bmod p$.

iii) 'A' sends R_1 to 'B' without sending value of x.

iv) 'B' send R_2 to 'A' without sending value of y.

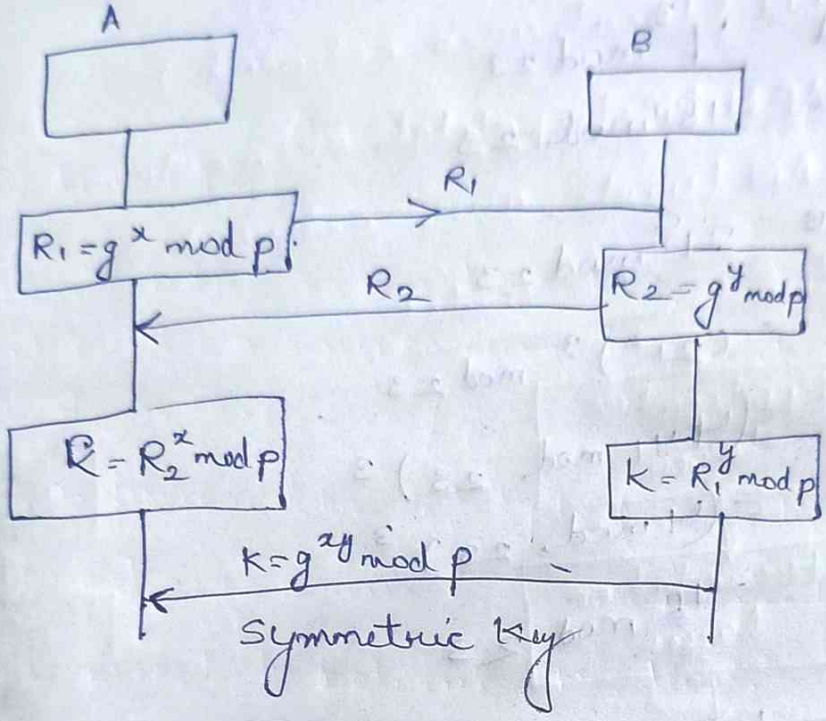
v) 'A' calculates $k = (R_2)^x \bmod p$

vi) B calculates $k = (R_1)^y \bmod p$

k is the symmetric key for the session.

$K = (g^y \bmod P)^x \bmod P = g^{xy} \bmod p$

$K = (g^x \bmod P)^y \bmod P = g^{xy} \bmod p$



Eg: Suppose A and B choose $g = 7$ & $P = 23$

1. A chooses $x = 3$

$$R_1 = 7^3 \bmod 23 = (7^2 \times 7) \bmod 23$$

$$= 3 \times 7 \bmod 23$$

$$= 21 \bmod 23$$

2. B chooses $y = 6$

$$R_2 = 7^6 \bmod 23$$

$$= (7^3)^2 \bmod 23$$

$$= 21^2 \bmod 23$$

$= 441 \bmod 23$

$= 4 \bmod 23$

$23 \overline{)441}\ 19$
23
211
207
4

$K_A = 4^3 \bmod 23$

$= 18 \bmod 23$

$K_B = 21^6 \bmod 23$

$= (21^2)^3 \bmod 23$

$= (441 \bmod 23)^3$

$= (4 \bmod 23)^3$

$= 18 \bmod 23$

$K = 7^{3 \times 6} \bmod 23$

$= 7^{18} \bmod 23$

$= 18 \bmod 23$

06/11/26 Computer network

5 components

i) Sender

ii) Receiver

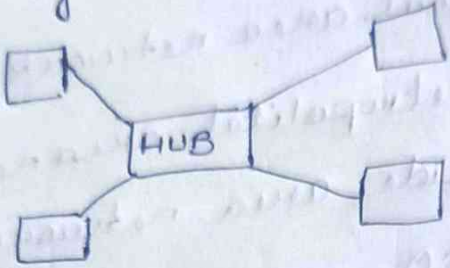
iii) message

iv) communication media

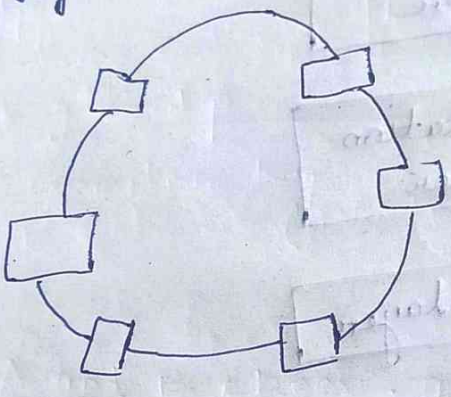
v) Protocol IP, TCP/IP, HTTP

Topology -
The structure of interconnection.

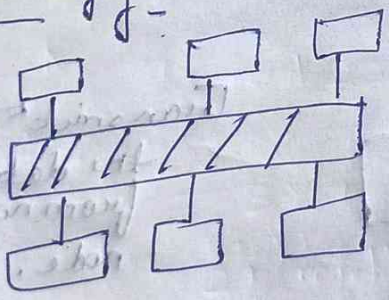
1. Star topology -



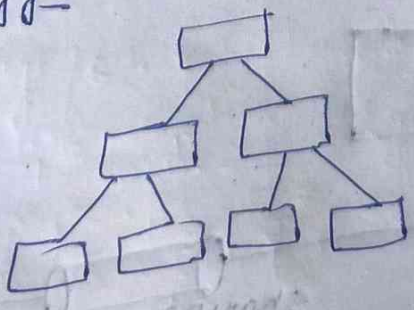
2. Ring topology



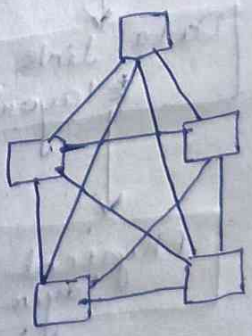
3. Bus topology -



4. Tree topology -



5. Mesh topology



Types of network

1) PAN (Personal Area Network)

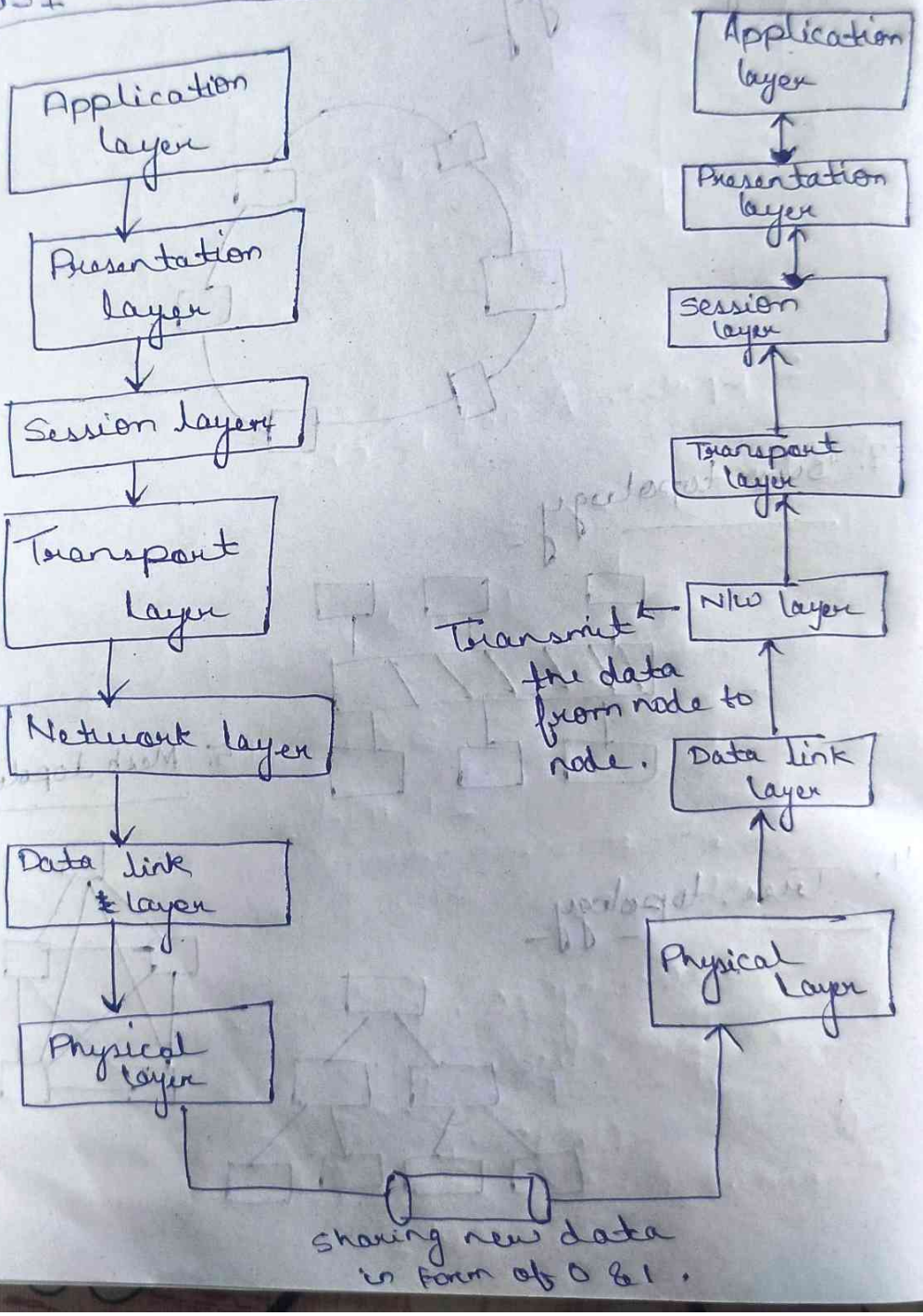
Eg:- Bluetooth

2) LAN (Local area network)

3) MAN (Metropolital area network)

4) WAN (wide area network)

OSI :- ISO model



TCP / IP protocol suite

